

CE 502: Groundwater Flow and Contaminant Transport

Course Code: CE 502

Course Name: Groundwater Flow and Contaminant Transport

L-T-P-C: 3-0-0-3

Prerequisite: Water Resources Engineering, Programming knowledge, Faculty Approval

Students intended for: UG/PG

Elective or Core: Elective

Approval: 9th Senate

Course content

- **Introduction**

Hydrologic cycle, Subsurface flow and basics, Darcy's law, emphasizing the role of groundwater in the hydrologic cycle, different types of aquifer and flow properties, types and sources of contamination. [2 Lectures]

- **Flow through Porous Medium**

Flow through filters and soil columns, fractured rock and stratified porous medium, Well Hydraulics. [2 Lectures]

- **Groundwater Flow**

Hydraulic Potential, the Steady-state Groundwater flow equation, derivations of flow equations for homogeneous and heterogeneous porous medium, application of Darcy's law, Flow Modelling. [4 Lectures]

- **Contaminant Transport**

Groundwater Pollution, Advection and Dispersion, Sorption and Diffusive Mass Transfer, Aquifer Remediation, contaminant transport model in homogeneous and heterogeneous porous medium, Flow parameters identification and estimation. [6 Lectures]

- **Groundwater Flow and Transport modeling**

Analytical and Numerical methods, solution of flow equation and ADE, MIM, MPNE. [6 Lectures]

- **Field application of Transport equations**

Use of numerical model for simulation experimental data and prediction spreading of contaminant at field scale. [6 Lectures]

Groundwater Lab

1. Experiments for reactive and nonreactive solute transport through porous media
2. Estimation of equilibrium sorption coefficients of various reactive solutes using linear and nonlinear isotherms
3. Inverse problem for source identification
4. Rainfall runoff simulation for surface and groundwater interaction

Text Books

1. Zheng, Chunmiao, and Gordon D. Bennett, **Applied contaminant transport modeling**, Vol. 2, Wiley-Interscience, 2002.
2. Yong, Raymond Nen, Abdel-Mohsen Onsy Mohamed, and Benno P. Warkentin, **Principles of contaminant transport in soils**, Elsevier Science Publishers, 1992.
3. Todd, David K., and Larry W. Mays, **Groundwater hydrology edition**, Wiley, New Jersey, 2005.
4. Freeze, R. Allan, John ARA Cherry, and Cherry JA, **Groundwater**, No. 556.3 FRE. 1979.

Reference Books:

1. Bear, Jacob, Chin-Fu Tsang, and Ghislain De Marsily, **Flow and contaminant transport in fractured rock**, Academic Press, 2012.
2. Grathwohl, Peter, **Diffusion in natural porous media**, Academic Press, 2012 Kluwer, 2002.
3. Dagan, Gedeon, **Flow and transport in porous formations**, Springer-Verlag GmbH & Co. KG., 1989.